

Sequential Logic Circuits - II

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Agenda

- ◆ Registers
- ◆ Counters

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Registers

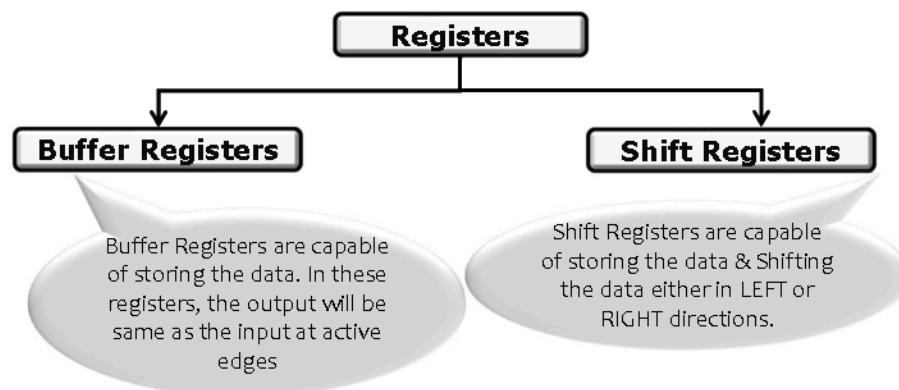
- ♦ Group of many flipflops to store multiple bits of information is called as “Register”.
- ♦ Registers can be used to perform the following two different functions.
 - ♦ Data storing
 - ♦ Data shifting

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Classification of Registers

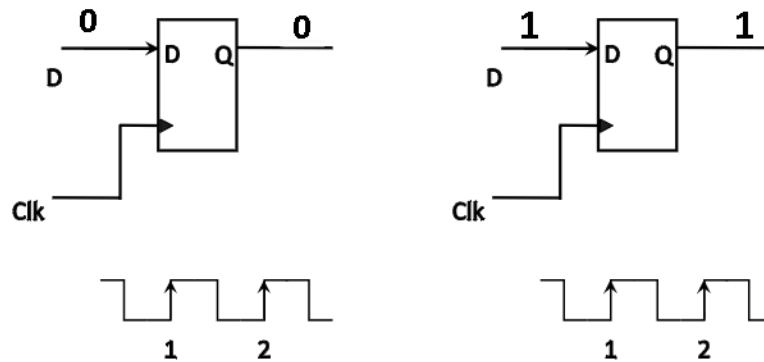
Registers can be classified as follows.



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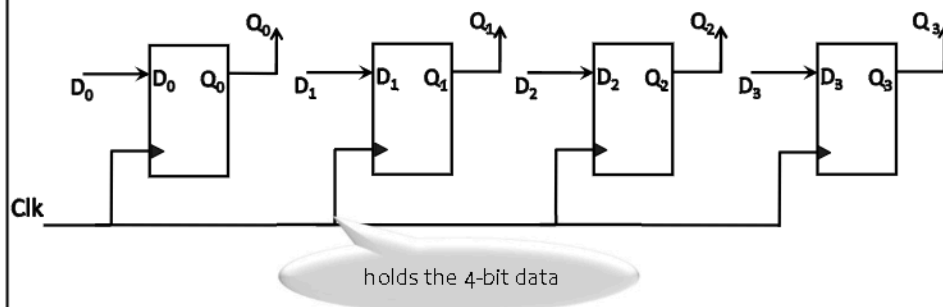
Functionality of a D flipflop



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Buffer Register

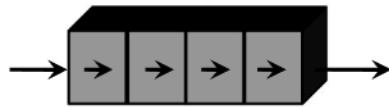


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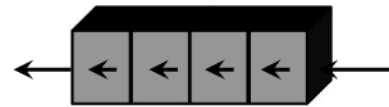
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Serial Shift Registers

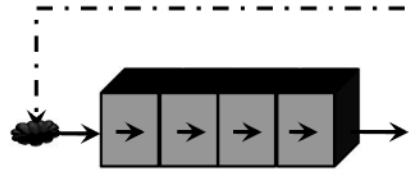
Serial In Serial Out (Right Shift)



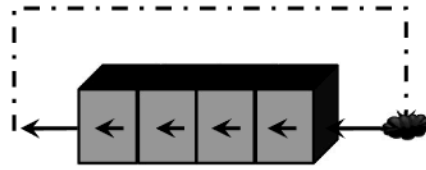
Serial In Serial Out (Left Shift)



Rotate Right



Rotate Left

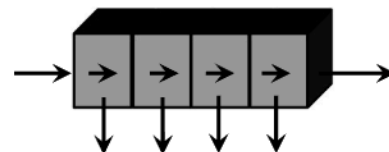


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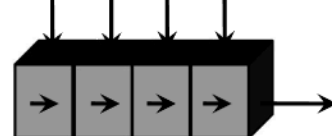
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Parallel Shift Registers

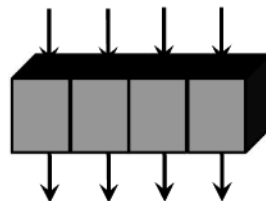
Serial In Parallel Out



Parallel In Serial Out



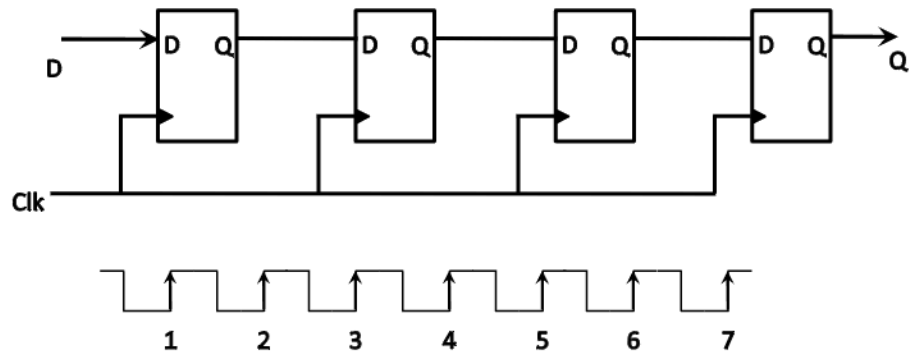
Parallel In Parallel Out



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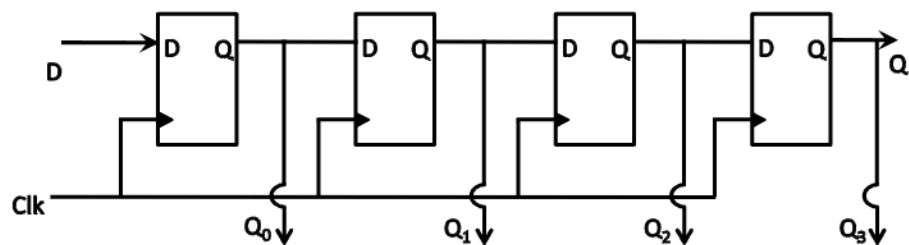
SISO Shift Registers



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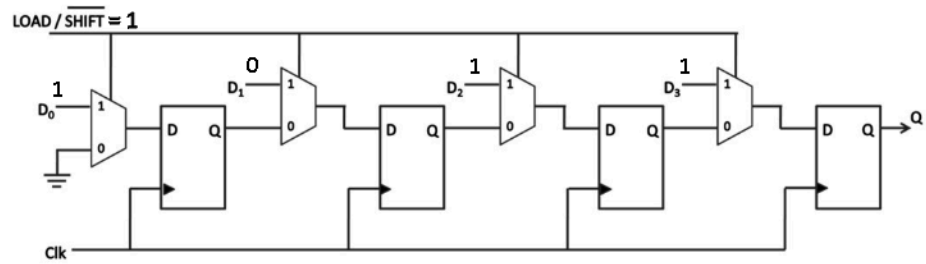
SIPO Shift Registers



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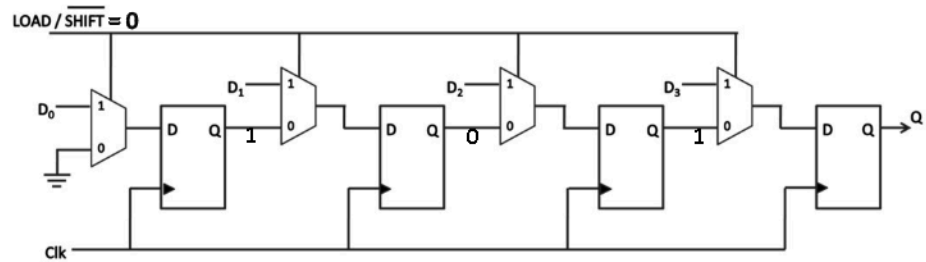
PISO Shift Register



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PISO Shift Register

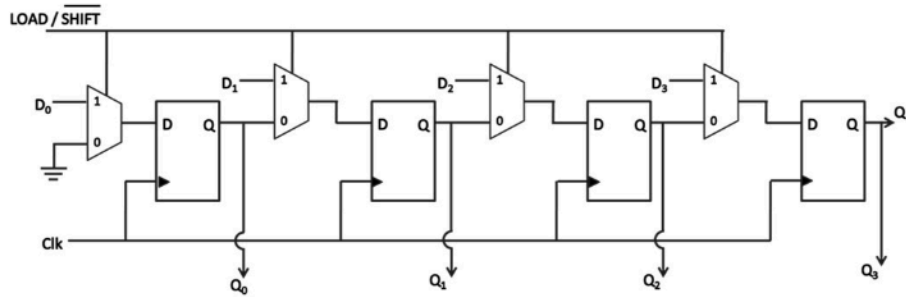


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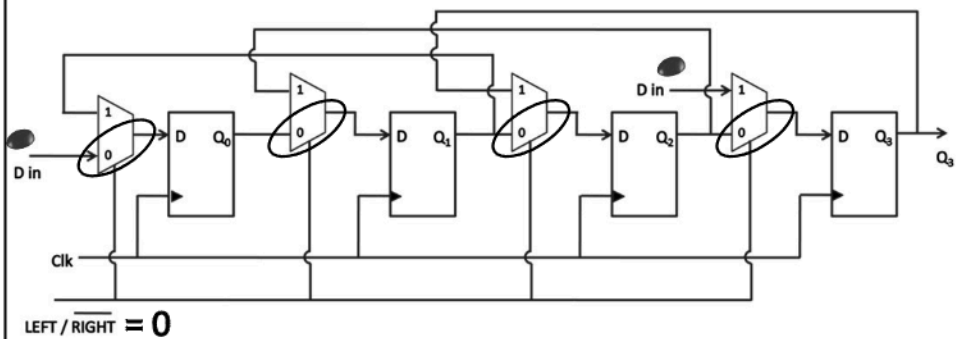
PIPO Shift Register



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Bidirectional Shift Registers

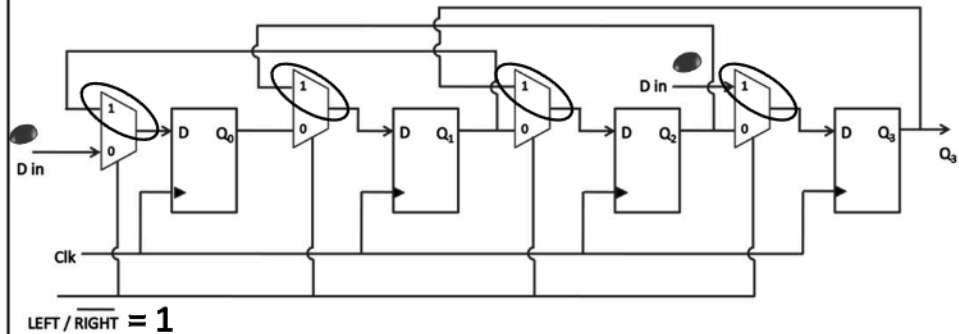


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Bidirectional Shift Registers



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Counters

What is a counter ?

- ♦ As the name suggests, it counts the no. of clock pulses.
- ♦ This counting can be done in a
 - ♦ Sequential manner to count the no. of clock pulses.
 - ♦ Random manner to generate a specific sequence.

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Classification of Counters

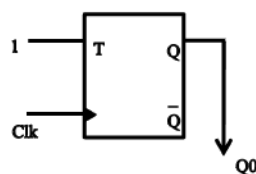
Counters can be classified based on

- ♦ **Clock** : Synchronous or Asynchronous
- ♦ **Clock Trigger** : Posedge trigger or Negedge trigger.
- ♦ **Counts** : Binary, Decade
- ♦ **Count Direction** : UP, DOWN, UP-DOWN, RANDOM
- ♦ **Flipflops Used** : JK or T or D.

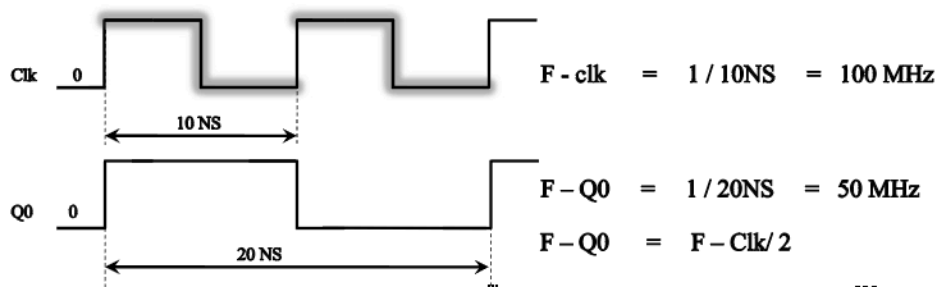
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Concept of Frequency Division



- ♦ The output of a +ve edge triggered T-flipflop toggles at every +ve edge in the clock signal.
- ♦ And remains constant during the period other than the instant of +ve edge of the clock.



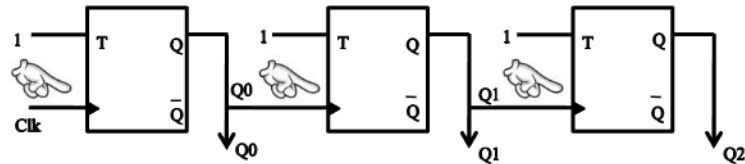
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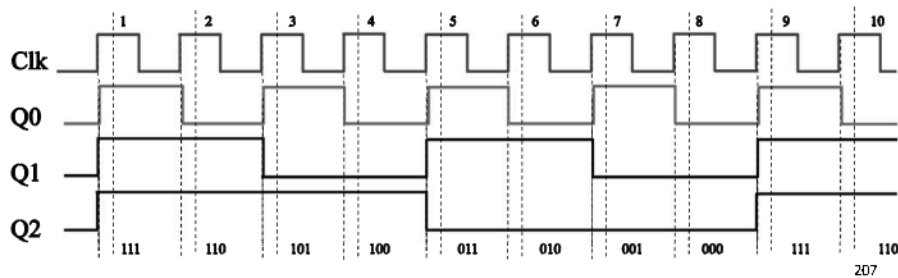
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Asynchronous counter

3 Bit Asyn Down Counter using +ve edge triggered FF



$$F(Q0) = F(\text{Clk}) / 2 \quad F(Q1) = F(Q0) / 2 \quad F(Q2) = F(Q1) / 2$$

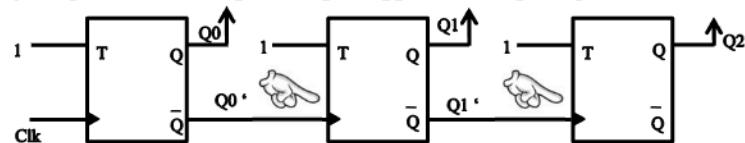


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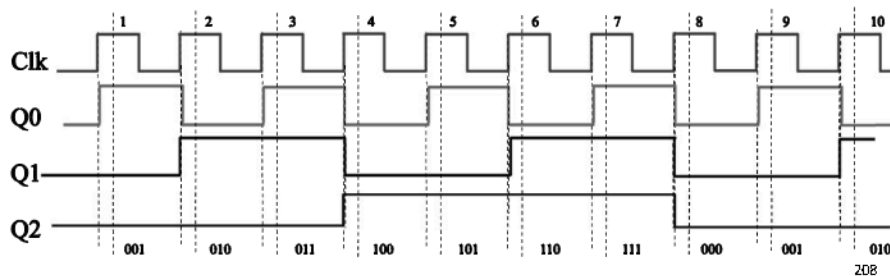
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Asynchronous counter

3-bit Asyn. up counter using Posedge triggered T-Flip Flop



$$F(Q0) = F(\text{Clk}) / 2 \quad F(Q1) = F(Q0) / 2 \quad F(Q2) = F(Q1) / 2$$



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[illegible]

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UP/DOWN Counter

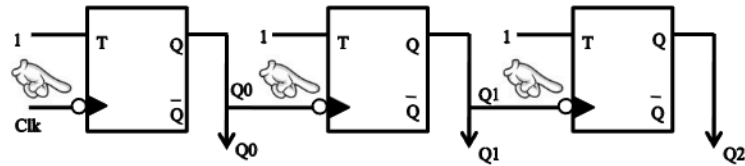
UP / DOWN = 1

The diagram illustrates a 4-bit UP/DOWN counter. It consists of four J-K flip-flops (labeled 0, 1, 2, 3) connected in a chain. The clock input (clk) is connected to the clock input of flip-flop 0. The output Q0 of flip-flop 0 is connected to the clock input of flip-flop 1. The output Q1 of flip-flop 1 is connected to the clock input of flip-flop 2. The output Q2 of flip-flop 2 is connected to the clock input of flip-flop 3. The output Q3 of flip-flop 3 is the final output. The UP/DOWN control signal is connected to the '0' input of each XOR gate. The '1' input of each XOR gate is connected to the output Q of the corresponding flip-flop. The output of each XOR gate is connected to the J input of the corresponding flip-flop. The K input of each flip-flop is connected to 1. The outputs are Q0, Q1, Q2, and Q3.

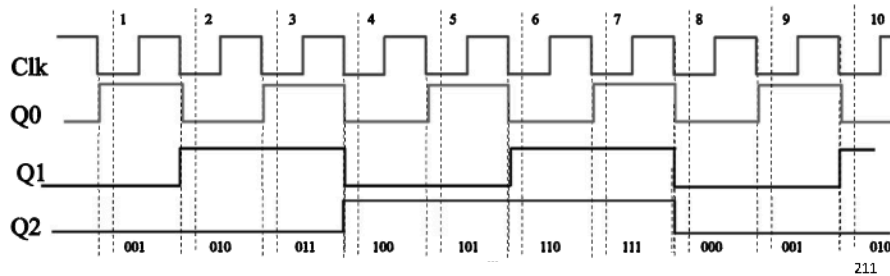
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Asynchronous counter

3-bit Asyn. up counter using Negedge triggered T-Flip Flop



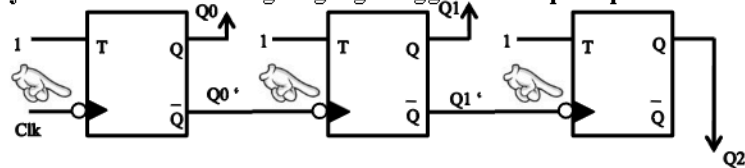
$$F(Q0) = F(\text{Clk}) / 2 \quad F(Q1) = F(Q0) / 2 \quad F(Q2) = F(Q1) / 2$$



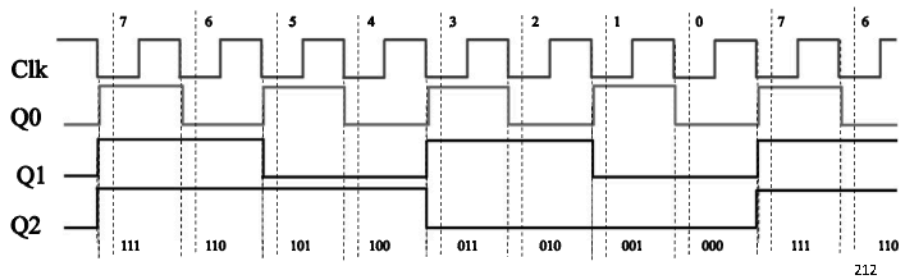
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Asynchronous counter

3-bit Asyn. down counter using Negedge triggered T-Flip Flop



$$F(Q0) = F(\text{Clk}) / 2 \quad F(Q1) = F(Q0) / 2 \quad F(Q2) = F(Q1) / 2$$



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Synchronous UP Counter

UP Counting Sequence

Q_2	Q_1	Q_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1
0	0	0

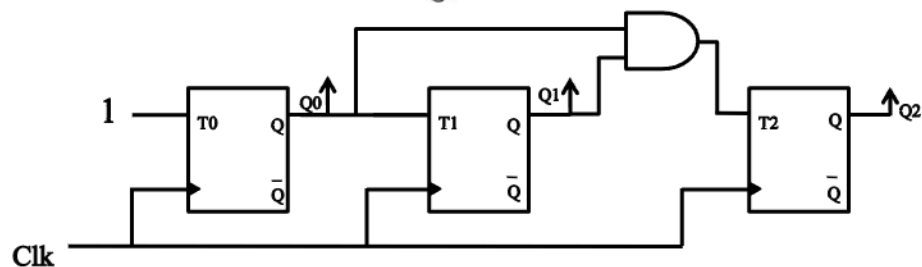
- ♦ Q_0 toggles for every cycle.
- ♦ And Q_1 toggles only when Q_0 is '1'
- ♦ Q_2 toggles only if (Q_1 & Q_0) are '1'.
- ♦ **Important Point :** T FF toggles only when the T input is '1'

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Synchronous UP Counter

- ♦ Q_0 toggles for every cycle.  ♦ So, T_0 should be supplied with '1'.
- ♦ Q_1 toggles when Q_0 is '1'  ♦ T_1 should be supplied with Q_0 .
- ♦ Q_2 toggles only if (Q_1 & Q_0) '1'  ♦ T_2 should be supplied with Q_0 & Q_1



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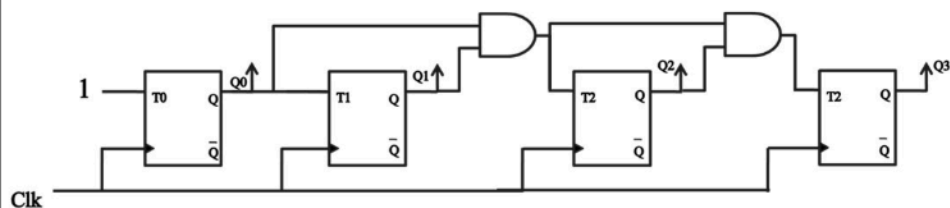
Modulus of a Counter

- ◆ The number of unique states that a counter may have before the sequence repeats itself is the modulus of the counter.
- ◆ An UP counter with 3 flipflops counts from 0 to 7 and again repeats from 0 after 7. So, the no. of states is 8 (i.e., 0 to 7).
- ◆ So, a counter with 'n' no. of flipflops counts from 0 to $2^n - 1$. So, the Modulus of this counter is 2^n .
- ◆ The output frequency of the last FF of any counter will be the clock frequency divided by the MOD of the counter.
- ◆ A T flipflop with T input tied to Vcc acts as divide by 2 counter.

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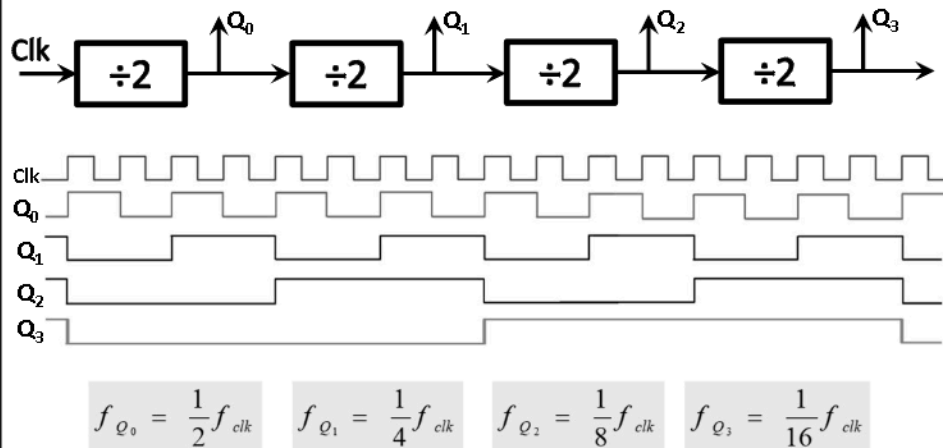
Mod-16 Counter



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Frequency Division using Counters



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Advantages of Syn. Counters over Asyn. Counters.

- ♦ The outputs of all the flipflops will change simultaneously.
- ♦ A synchronous counter can operate at much higher frequency, but the circuitry are more complex than that of the asynchronous counter.

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Counter Design

- ♦ **Step 1:** Derive the State Transition Diagram
- ♦ **Step 2:** Draw State Transition Table
- ♦ **Step 3:** Choose flip-flop type for implementation

Use Excitation Table to remap Next State Functions

- ♦ **Step 4:** Obtain the expressions for input variables of the flipflops in terms of the present state values.
- ♦ **Step 5:** Minimize them using K-maps
- ♦ **Step 6:** Implement Resulting Logic with the minimised expressions.

You will learn during the advanced course

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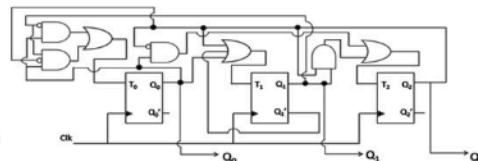
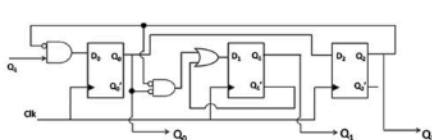
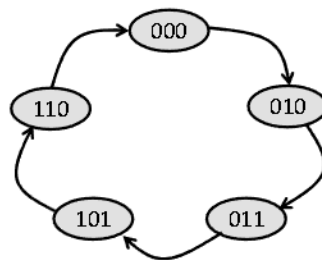
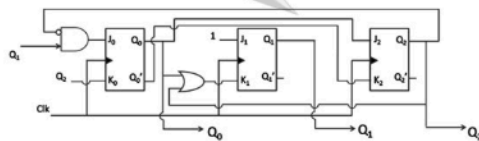
Counter Design

Design a counter that generates the sequence.

Example :

0,2,3,5,6,0,---

Counter using different ffs are shown here



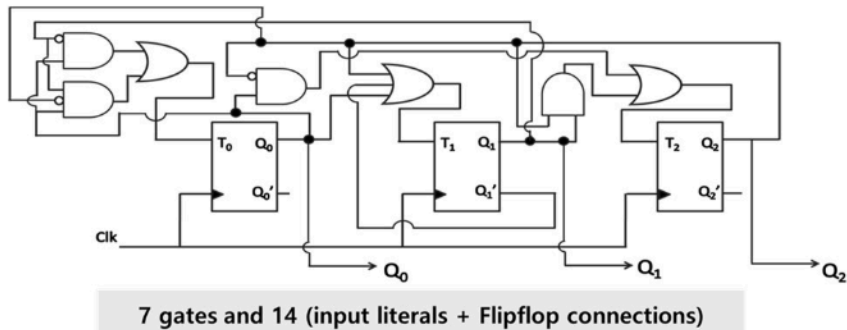
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Comparison

- ♦ T FFs well suited for straightforward binary counters
- ♦ But yielded worst gate and literal count for this example

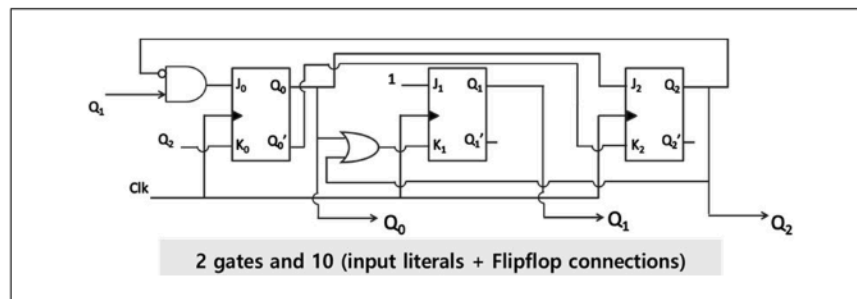


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Comparison

- ♦ JK FFs yielded lowest gate count
- ♦ Tend to yield best choice for packaged logic where gate count is key



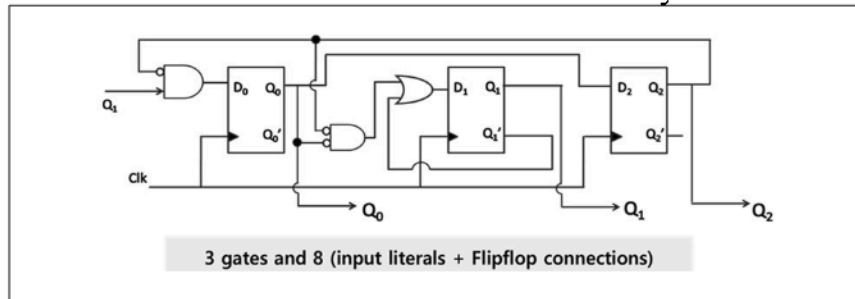
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Comparison

- ◆ D FFs yield simplest design procedure

- ◆ Best literal count
- ◆ D storage devices very much transistor efficient in VLSI
- ◆ Best choice where area/literal count is the key



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Shift Register Counter

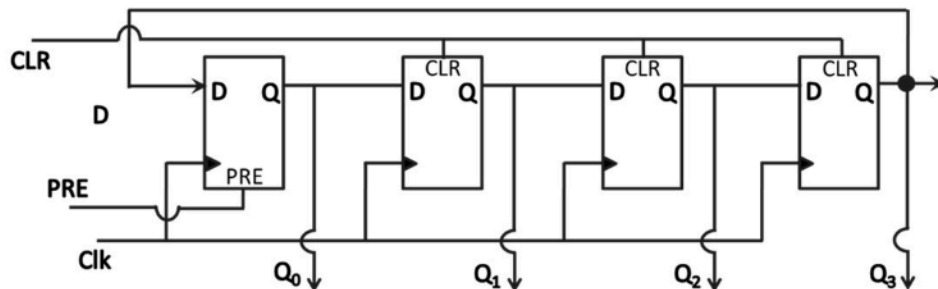
- ◆ We are aware of the fact that counters are used for producing a special sequence of numbers.
- ◆ Similarly, a shift register with the serial output connected back to the serial input to produce special sequence.
- ◆ Two of most common type of shift register counters are :
 - ◆ Ring counter
 - ◆ Johnson counter [Shift Counter / twisted ring counters].

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Ring Counter



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Ring Counter

Cycle Number	Q_0	Q_1	Q_2	Q_3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	0	0	0
5	0	1	0	0
6	0	0	1	0
7	0	0	0	1

4 Cycles

4 Cycles

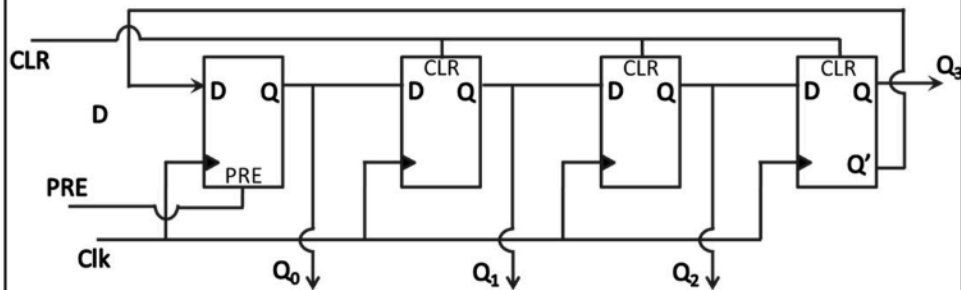
Modulus of a 'N' bit ring counter is 'N'.

For every 4 clock cycles, this counter goes into the same set of states. So, modulus of a 4 bit ring counter is 4.

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Twisted Ring Counter



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Twisted Ring Counter

Cycle Number	Q_0	Q_1	Q_2	Q_3	Q_3'
0	1	0	0	0	1
1	1	1	0	0	1
2	1	1	1	0	1
3	1	1	1	1	0
4	0	1	1	1	0
5	0	0	1	1	0
6	0	0	0	1	0
7	0	0	0	0	1
8	1	0	0	0	1

For every 8 clock cycles, this counter goes into the same set of states. So, modulus of a 4 bit Johnson counter is 8.

So, modulus of a 'N' bit Johnson counter is '2N'.

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